Can an AI Explanation Be Misused? Testing Whether SHAP and LIME Help Patients Game a Clinical Diabetes Risk Model

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Abstract

Explainable AI (XAI) methods such as SHAP and LIME are increasingly required by regulation to accompany machine learning predictions in healthcare. A concern is that feature attributions may also serve as instructions for gaming the model—telling a patient which self-reported inputs to change for a better risk score—so that explanations should be withheld. But a patient can also simply try different answers and watch the score move. We compare these two strategies on a diabetes risk model trained on NHANES public health survey data, where height and weight are recorded both as self-reported and as measured by a nurse while the diabetes label comes from a blood test (HbA1c). Lies are restricted to the size and direction of real self-reporting errors observed in NHANES. We train four models—Random Forest and Logistic Regression, each with nurse-measured or self-reported height/weight—and measure the *explanation equivalence* N^*: how many blind attempts give the same mean score reduction as one explanation-guided lie. In our pre-registered evaluation on the held-out 2017–2018 NHANES cycle ($n = 887$ test patients), $N^* = 0.0$ (95% CI 0.0–0.0) in all eight configurations: one explanation-guided lie lowers the risk score *less* on average than one random realistic lie (e.g. 0.013 vs. 0.024 for Random Forest with SHAP). Explanations often point the patient away from a profitable lie, or, for the Random Forest, towards a lie that raises the score. The model that uses nurse-measured inputs cannot be gamed at all. Withholding explanations is therefore not an effective safeguard; verifying self-reported inputs is. We release our code, data pipeline, and pre-registered analysis plan.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Social and professional topics** → *Medical records.*

Keywords

explainability, SHAP, LIME, adversarial, gaming, diabetes, NHANES, clinical AI, transparency, fairness

1 Introduction

The deployment of machine learning models in high-stakes clinical settings has prompted widespread calls for transparency. Regulations such as the EU AI Act and GDPR's "right to explanation" push developers to provide explanations alongside predictions [7]. In diabetes risk prediction, a model might inform a patient: "Your risk score is 0.72. Your weight is the most important contributing factor."

This transparency creates a potential dual-use risk: the same explanation that helps a clinician understand the model may help a patient *game* it by adjusting what they self-report. A patient told that weight is the top risk factor might under-report their weight at the next visit. If explanations act as instructions for cheating, one response is to hide them.

Our Question. A patient does not need an explanation to game a model: they can try a few different answers and keep the one with the best score. The relevant question is therefore not whether an explanation helps at all, but whether it helps *more than ordinary trial-and-error*. If explanations add little over a few blind attempts, restricting them is not an effective safeguard. If they add a lot, the benefit of transparency must be weighed against the gaming risk.

Contributions.

(1) We introduce *explanation equivalence* (N^*): the number of blind attempts required to achieve the same mean score reduction as one explanation-guided attempt.
(2) We bound the adversary's lies by *real* self-reporting behaviour: NHANES records both self-reported and nurse-measured height and weight, so we take the size and direction of a realistic lie from the observed gap.
(3) We train two models that differ *only* in whether height/weight are self-reported or measured, isolating the effect of input verification.
(4) We conduct a *pre-registered* evaluation of SHAP and LIME on held-out data and find that, under realistic lies, one explanation is worth fewer than one blind attempt ($N^* = 0$ in all configurations).

2 Related Work

Explainability Methods. Local explanation methods such as LIME [14] and SHAP [11] produce feature attributions that identify the most important inputs for a given prediction. These methods are increasingly embedded in clinical decision support systems [1]. Explanations can look convincing yet be disconnected from the model [2], and Bilodeau et al. [4] proved that for broad model classes feature attributions can do no better than random guessing at predicting how the model responds to a change in its input—showing that a problem is possible, but not how large it is in practice.

Adversarial XAI.. Slack et al. [16] demonstrated that SHAP and LIME can be manipulated at the *model* level to produce misleading explanations, and Bordt et al. [5] argue that post-hoc explanations suit legal transparency rules poorly because providers can manipulate them. Our work is complementary: we study gaming at the *user* level, asking whether an honest model's explanations help a user game their score. Milli et al. [12] showed that a few explanation queries can replace many ordinary queries for reconstructing a model, and Shokri et al. [15] showed that explanations can leak

private training data; both concern the model or its data rather than a patient changing their own answers.

Strategic Classification. Ghalme et al. [6] found that hiding the decision rule usually does not help the decision-maker, and Hikri and Rosenfeld [8] extended this to partial disclosure. König et al. [10] note that advice can improve a prediction without improving the person, just as lying about weight is not losing weight. Unlike these works, we bound manipulation by real human misreporting and compare explanations against an equal-cost blind baseline.

Self-Reporting Bias. NHANES [9] systematically collects both self-reported and nurse-measured anthropometrics, enabling a direct empirical measurement of how people misreport.

3 Problem Setup

3.1 Threat Model

We model an adversary (patient) who:

- Receives a diabetes risk score $f(\mathbf{x}) \in [0, 1]$ from a clinical AI system and wishes to lower it.
- Can change only *self-reported* features $G = \{\text{height}, \text{weight}\}$; nurse-measured values and lab results cannot be changed.
- Lies only *realistically*: in the direction people actually misreport, by at most the lie budget $B(g)$.
- Either receives one SHAP/LIME explanation, or receives none and makes N blind attempts, keeping the best score.

3.2 Realistic Lies

For each gameable feature g, the lie budget is

$$B(g) = \text{Percentile}_{95}\big(|x_g^{\text{self}} - x_g^{\text{measured}}|\big),$$

computed on the 2015–2016 development cycle: $B(\text{height}) = 7.28\,\text{cm}$ and $B(\text{weight}) = 8.96\,\text{kg}$. People over-report height (mean $+1.62\,\text{cm}$) and under-report weight (mean $-0.87\,\text{kg}$), so a realistic lie *increases* height or *decreases* weight. Each lie has a size drawn uniformly from $(0, B(g)]$; the same rule is used by both strategies, so the explanation contributes only *information* (which feature, which direction), never a larger lie.

3.3 Explanation Equivalence

Definition 3.1 (Explanation Equivalence N^*). Let R_{exp} be the score reduction from one explanation-guided lie, averaged over patients, and let $R_{\text{blind}}(N)$ be the reduction from the best of N blind lies, averaged over patients. The *explanation equivalence* is

$$N^* = \inf\{N \geq 0 : R_{\text{blind}}(N) \geq R_{\text{exp}}\},$$

estimated by linear interpolation on $N \mapsto R_{\text{blind}}(N)$ for

$$N \in \{1, 2, 3, 5, 10, 15, 20, 30, 50, 100\}.$$

$N^* = 0$ if one explanation does no better than a single blind attempt; $N^* = 101$ if it beats 100 attempts (censored).

This is the pre-registered primary metric: the average is taken over patients *before* interpolating. As secondary metrics we also report the per-patient N^* distribution.

4 Methodology

4.1 Dataset

We use the NHANES public health survey [9] and merge the demographics (DEMO), body measures (BMX), weight history questionnaire (WHQ), glycohaemoglobin (GHB), blood pressure (BPX), smoking (SMQ), and physical activity (PAQ) files for the 2015–2016 (development) and 2017–2018 (held-out test) cycles. Refused and don't-know codes are removed per variable as in the NHANES codebooks. Retaining adults (≥18 years) with complete records for both feature sets gives 4,889 participants in 2015–2016 (12.4% with diabetes, HbA1c ≥ 6.5% [3]) and 4,431 in 2017–2018 (13.6%). Within each cycle we use a stratified 80/20 split; the final evaluation uses the $n = 887$ test patients of the 2017–2018 cycle, which was not examined during development.

4.2 Models

Each model uses seven features: age, sex, systolic blood pressure, smoking (ever 100 cigarettes), vigorous recreational activity, height, and weight. *Model A* uses nurse-measured height/weight; *Model B* uses the self-reported values; everything else is identical. We train a Random Forest (RF; 100 trees, maximum depth 10) and a Logistic Regression (LR; standardised inputs) for each, with balanced class weights to account for the 13% prevalence (table 1). On the 2017–2018 test set the models reach AUC 0.77–0.80 and recall 0.69–0.78; Models A and B perform almost identically.

4.3 Explanations

SHAP uses TreeExplainer for RF and LinearExplainer for LR (background: a seeded random sample of 100 training rows). LIME uses its tabular explainer with 500 samples, its default discretisation of continuous features, and all seven features shown.

4.4 Adversarial Strategies

Strategy 1 (Explanation-Guided). The patient requests one explanation and reads it the way it is presented: features ranked by importance, each pushing the risk up or down. Going down the ranking, the patient takes the first gameable feature whose risk-lowering direction is a realistic lie (SHAP: lower a feature with positive attribution, raise one with negative attribution; LIME: leave a risk-raising bin, e.g. raise the feature if the explanation says "*weight ≤ 62 kg* raises risk"). If no realistic lie is suggested, the patient does not lie. Otherwise they lie once, with a size drawn as in §3.2.

Strategy 2 (Blind Trial-and-Error). The patient makes N sequential attempts. Each picks a gameable feature at random and lies in its realistic direction with a size drawn as in §3.2; the best score so far is kept. The simulation runs once up to $N = 100$ and records the running best at each milestone, so blind performance at large N is not over-estimated.

Every patient has its own fixed random seed, so the SHAP and LIME strategies, and Models A and B, are compared against identical blind attempts.

Table 1: Model Performance on NHANES 2017–2018 Test Set

Config	Input	Model	Accuracy	Precision	Recall	F1	AUC-ROC
measured_rf	measured	RF	0.7396	0.3007	0.686	0.4181	0.7981
measured_lr	measured	LR	0.6652	0.2569	0.7686	0.3851	0.7734
reported_rf	reported	RF	0.7407	0.3032	0.6942	0.4221	0.7933
reported_lr	reported	LR	0.6685	0.2604	0.7769	0.39	0.7796

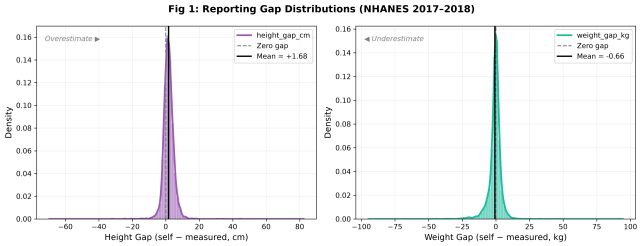


Figure 1: Self-reported minus nurse-measured height and weight (NHANES 2017–2018). Height is over-reported and weight under-reported.

4.5 Statistical Analysis

For each configuration we report N^* with a 95% CI from 1,000 bootstrap resamples of patients; a one-sided paired Wilcoxon signed-rank test [17] of the per-patient explanation reduction against the best of $N = 10$ blind attempts (H_1: explanation > blind); and Cohen's d. All models were implemented with scikit-learn [13].

4.6 Pre-Registration and Deviations

The analysis plan (hypotheses, metric, tests, interpretation thresholds) was recorded as the git tag `v0.1-preregistration` in the project repository before the 2017–2018 data was processed. Before the final run reported here, we corrected implementation errors so that the code matches that plan (primary N^* computed from the averaged curve; the pre-registered 2015–2016 lie budget; only self-reported values gameable). We also made six changes that *deviate* from it: lies only in the realistic direction; the same lie-size rule for both strategies (previously the explanation always used the maximum lie); LIME showing all seven features (previously five); balanced class weights; a random SHAP background; and per-patient random seeds. Each is a documented switch in the code and can be reverted for sensitivity analysis.

5 Results

5.1 Reporting Gap

Figure 1 shows self-reported minus nurse-measured values. In 2017–2018, participants over-report height by +1.69 cm on average (SD 3.23) and under-report weight by −0.67 kg (SD 4.84), closely matching 2015–2016 (+1.62 cm, SD 3.41; −0.87 kg, SD 4.79). The 95th-percentile gap in 2017–2018 (6.88 cm, 9.32 kg) is within 0.4 cm and 0.4 kg of the development budget we use.

5.2 Main Result: Explanation Equivalence

Table 2 and fig. 2 show that $N^* = 0.0$ (95% CI 0.0–0.0) in all eight configurations. For Model B, one explanation-guided lie lowers the risk score by 0.012–0.022 on average, while a single blind attempt already lowers it by 0.024–0.025, and the best of 10 blind attempts by 0.046–0.064 (table 3). The trial run on 2015–2016 (200 patients) gave the same picture ($N^* = 0.0$ in seven configurations and 1.0 for LR+LIME).

5.3 Why the Explanation Does Not Help

Two mechanisms explain the result (table 3). *First, the explanation often suggests no profitable lie.* A feature attribution describes a feature's contribution relative to an average patient, not which change would lower the score. For LR, SHAP gives $\beta_g(x_g - \mathbb{E}[x_g])$: a patient with below-average weight sees "weight lowers your risk" and is not pointed to weight, even though under-reporting weight would lower the score further. For 31% of patients (LR+SHAP) the explanation suggests no realistic lie at all; blind attempts lie anyway and, since realistic lies usually help, gain. *Second, for the Random Forest the suggested lie often backfires*: 23% (SHAP) and 31% (LIME) of explanation-guided lies *raise* the score, because local attributions do not describe the model's response over the whole range of a lie. This is the practical face of the impossibility result of Bilodeau et al. [4].

5.4 Per-Patient Distribution

Per patient (fig. 3), the explanation is no better than one blind attempt for 76.0% of patients (RF+SHAP) and worth 10 or more attempts for 8.7% (6.9% beat all 100 attempts). The per-patient mean N^* (7.7 for RF+SHAP, 2.6 for LR+SHAP) is therefore driven by a small tail, while the average patient gains more from a single blind attempt than from the explanation.

5.5 Model A vs. Model B

Model A, which uses nurse-measured height and weight, cannot be gamed: no strategy changes its score (fig. 4, table 4). Model B, which differs only in using self-reported values, can be lowered by 0.046–0.064 with ten blind attempts and 0.053–0.079 with a hundred. Both models have the same accuracy, so verifying the two self-reported inputs removes the vulnerability at no cost in performance.

5.6 SHAP vs. LIME Agreement

Because each patient receives the same lie size under SHAP and LIME, differences come from the explanation alone (fig. 5). The two methods lead to the same outcome for 53% (RF) and 40% (LR) of patients, with per-patient correlation $r = 0.60$ and 0.55. Both give

Table 2: Main Results: Explanation Equivalence N^* Across All 8 Configurations

Input	Model	Expl.	N patients	N^*	95% CI	Per-patient mean N^*	Per-patient median N^*	% Useless	% Worth 10+
Reported	RF	SHAP	887	0.0	[0.00, 0.00]	7.69	0.0	76.0%	8.7%
Reported	RF	LIME	887	0.0	[0.00, 0.00]	7.07	0.0	73.8%	8.1%
Reported	LR	SHAP	887	0.0	[0.00, 0.00]	2.58	0.0	66.4%	5.4%
Reported	LR	LIME	887	0.0	[0.00, 0.00]	2.99	0.0	56.3%	6.1%
Measured	RF	SHAP	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	RF	LIME	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	LR	SHAP	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	LR	LIME	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%

Table 3: Mean risk-score reduction (0-1 scale): one explanation-guided lie vs. the best of N blind lies, and how often the explanation suggested no lie or a lie that raised the risk

Input	Model	Expl.	1 explanation	1 blind	10 blind	100 blind	% no lie suggested	% lie raised risk
Reported	RF	SHAP	0.0129	0.024	0.0638	0.0785	19.3%	23.0%
Reported	RF	LIME	0.0117	0.024	0.0638	0.0785	3.3%	31.1%
Reported	LR	SHAP	0.0176	0.0248	0.0457	0.0525	31.1%	0.0%
Reported	LR	LIME	0.0223	0.0248	0.0457	0.0525	7.7%	0.0%
Measured	RF	SHAP	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	RF	LIME	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	LR	SHAP	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	LR	LIME	0.0	0.0	0.0	0.0	100.0%	0.0%

Table 4: Model A (Measured) vs Model B (Self-Reported): Vulnerability Comparison

Config	Model A N^*	Model B N^*	A: reduction, 10 blind	B: reduction, 1 explanation	B: reduction, 10 blind	B can be gamed
RF+SHAP	0.0	0.0	0.0	0.0129	0.0638	Yes
RF+LIME	0.0	0.0	0.0	0.0117	0.0638	Yes
LR+SHAP	0.0	0.0	0.0	0.0176	0.0457	Yes
LR+LIME	0.0	0.0	0.0	0.0223	0.0457	Yes

$N^* = 0$; LIME is slightly better than SHAP for LR (0.022 vs. 0.018) and slightly worse for RF (0.012 vs. 0.013).

5.7 Statistical Tests

The pre-registered one-sided Wilcoxon test of explanation > blind at $N = 10$ is not significant in any configuration ($p = 1.0$; table 5). The effect sizes are large and *negative* ($d = -0.79$ to -1.34 for Model B): ten blind attempts lower the score clearly more than one explanation.

Table 5: Statistical Tests: Explanation vs Blind (N=10), Wilcoxon Signed-Rank

Config	Wilcoxon p	Cohen's d	Significant
reported_rf_shap	1.000000	-0.801	No
reported_rf_lime	1.000000	-0.789	No
reported_lr_shap	1.000000	-1.336	No
reported_lr_lime	1.000000	-1.238	No
measured_rf_shap	1.000000	0.000	No
measured_rf_lime	1.000000	0.000	No
measured_lr_shap	1.000000	0.000	No
measured_lr_lime	1.000000	0.000	No

6 Discussion

6.1 Policy Implications

Our pre-registered interpretation guide set three bands: $N^* < 5$ (withholding explanations is not a meaningful safeguard), $5 \leq N^* \leq 20$ (moderate advantage), and $N^* > 20$ (explanations are powerful attack vectors). With $N^* = 0$ in every configuration, the result falls in the first band. Under realistic lies, an explanation does not give a patient a better route to a lower score than simply trying an answer; withholding it would remove information from honest users while barely slowing a motivated one. The vulnerability lies in the self-reported inputs: the measured-input model cannot be gamed, and the two models are equally accurate.

This conclusion runs against the intuition that explanations are "instructions for cheating". The reason is that a feature attribution answers "what contributed to this score?", not "what should I change?", and the two can point in different directions.

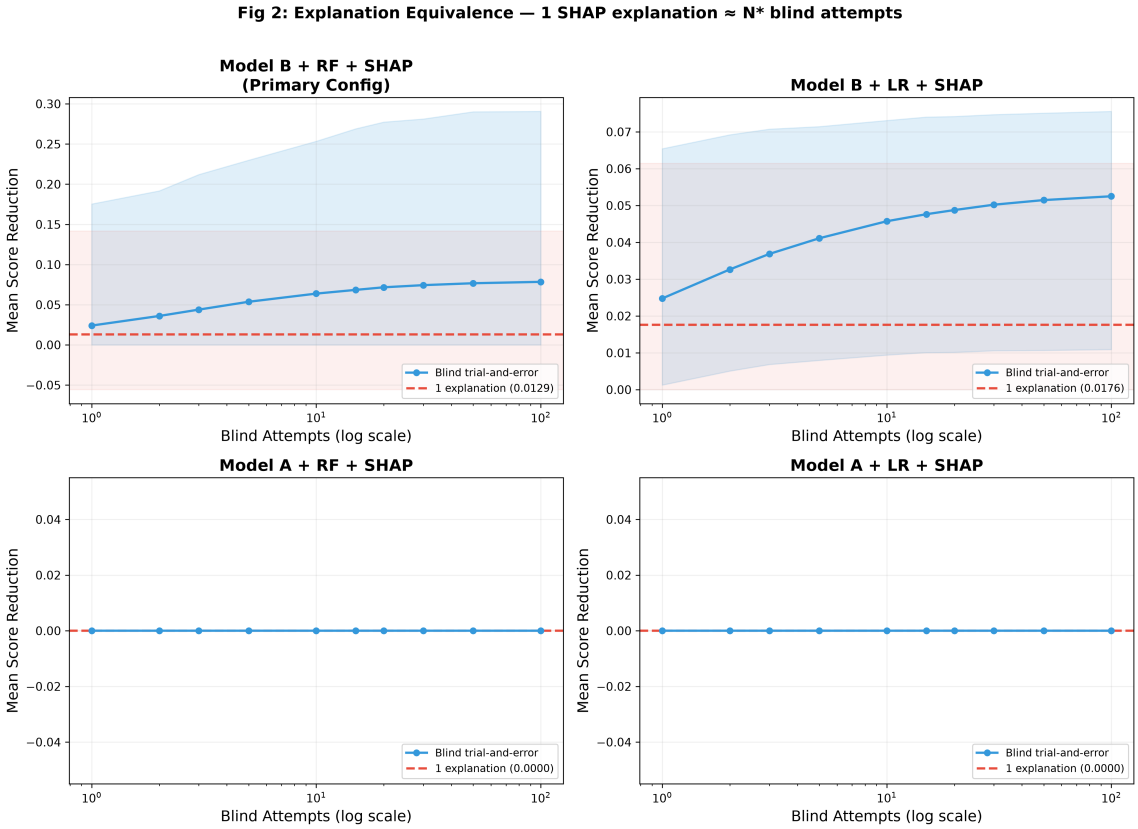


Figure 2: Mean score reduction from the best of N blind attempts (blue) vs. one explanation-guided lie (red dashed), SHAP. For Model B the explanation line lies below the first blind attempt, so $N^* = 0$. Model A cannot be gamed.

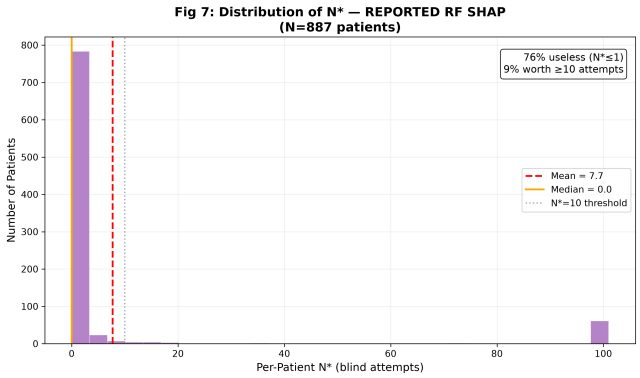


Figure 3: Distribution of per-patient N^* (Model B, RF + SHAP; $n = 887$ patients).

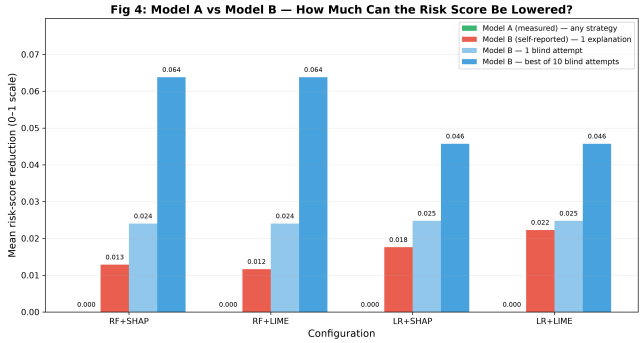


Figure 4: Mean risk-score reduction for Model A (nurse-measured inputs) and Model B (self-reported inputs).

6.2 Limitations

- **Model of the patient.** We model one way of reading an explanation (follow the top realistic suggestion once). A more sophisticated reader, or a counterfactual explanation, could extract more.
- **Single lie, single feature.** Each explanation-guided attempt changes one feature once; multi-feature lies could be more powerful for both strategies.
- **Lie model.** Lie size is uniform within the 95th-percentile gap and in the observed direction. Using the maximum lie, or allowing either direction, is a one-line change in our code; we leave that sensitivity analysis to future work.

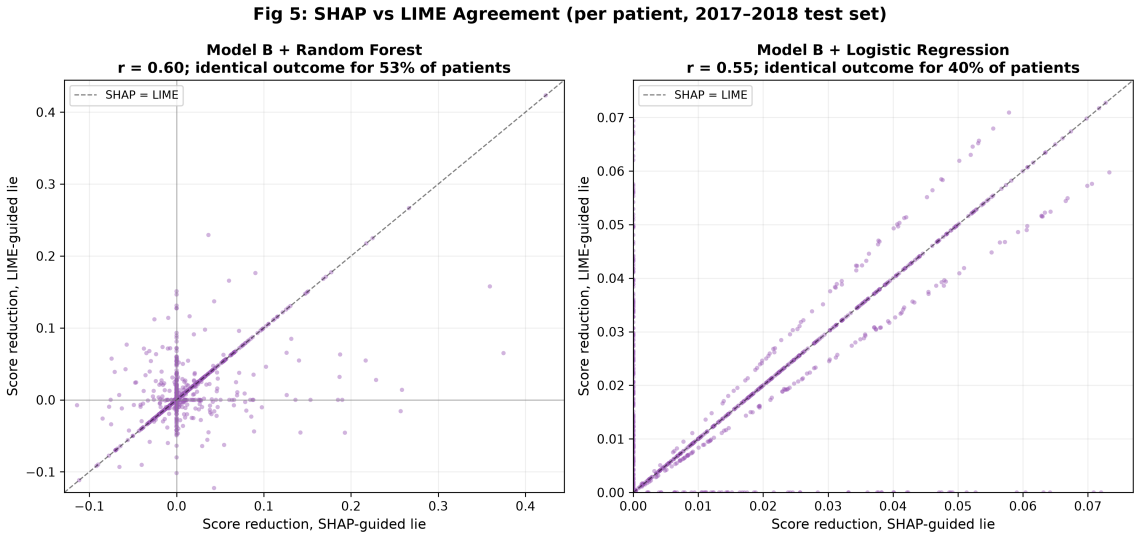


Figure 5: Per-patient score reduction from the SHAP-guided lie vs. the LIME-guided lie (Model B, 2017–2018 test set).

- **Single domain and dataset.** Results are for diabetes risk from anthropometric self-report in NHANES.
- **Pre-registration and deviations.** The plan was recorded as a git tag in the project repository rather than an independent registry, and six settings deviate from it (section 4.6).

6.3 Recommendations for Practitioners

(1) Prefer **verified inputs** (e.g. nurse measurement) over self-report for the features that drive high-stakes predictions.
(2) Do not withhold explanations as an anti-gaming measure; audit the gain they give with the N^* framework.
(3) Be aware that attributions are not advice: present them as "contributions" rather than "what to change".

7 Conclusion

We introduced the *explanation equivalence* metric N^* and applied it to a diabetes risk model using real NHANES survey data, with lies bounded by how people actually misreport. In a pre-registered evaluation on held-out 2017–2018 data, one SHAP or LIME explanation was worth fewer than one blind attempt ($N^* = 0$, 95% CI 0–0, in all eight configurations): explanations often pointed patients away from a profitable lie or, for the Random Forest, towards one that raised their score. A model using nurse-measured inputs could not be gamed at all. The effective safeguard against gaming is therefore verifying self-reported inputs, not withholding explanations.

Future Work. Sensitivity to the lie model, multi-feature and multi-step attacks, counterfactual explanations, other clinical domains, and a user study of how real patients read explanations.

Code and Data. Code, analysis plan and results: https://github.com/AkkiNikumbh/BTECH_PROJECT. NHANES data are public and are downloaded by the pipeline.

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